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25 April 1983
SUPERSEDING
MIL-S-17713B
12 March 1964

MILITARY SPECIFICATION

STEEL STRIP, CARBON, SPRING: FOR MECHANICAL TIME FUZE COMPONENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers carbon spring steel in strip form (see 6.1).

1.2 Classification.

1.2.1 Composition. The steel is furnished in the following compositions, as specified (see 3.1 and 6.2).

Composition 1 - Medium carbon spring steel, low phosphorus, low sulfur.
Composition 2 - Regular medium carbon spring steel.
Composition 3 - High carbon spring steel.

1.2.2 Condition. The steel is furnished in the following conditions:

Composition 1 - Soft type annealed - dead soft.
Composition 2 - Soft type annealed - dead soft.
Composition 3 - Spheroidize annealed type.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

Fed. Std. No. 66 - Steel: Chemical Composition and Hardenability.
Fed. Test Method Std. No. 151 - Metals; Test Methods.
Fed. Std. No. 183 - Continuous Identification Marking of Iron and Steel Products

MILITARY

MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E 290 - Semi-Guided Bend Test for Ductility of Metallic Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Pilot sample. Unless otherwise specified in the invitation for bids, contract or order (see 6.2), the supplier shall submit a sample, prior to the shipment of each lot, for blanking, forming, and piercing to the facility designated in the contract or order. Pilot samples from the same supplier may not be required if the accepted orders are within a three-month period.

3.2 Chemical composition. The chemical composition of the steel shall meet the requirements of table I. A cast or heat analysis of each heat of steel shall be furnished by the contractor.

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TABLE I. Chemical composition, cast or heat analysis

Composition	Carbon percent	Manganese percent	Phosphorus, maximum percent	Sulphur, maximum percent	Silicon percent
1	0.65-0.75	0.30-0.60	0.025	0.025	0.15-0.25
2	0.65-0.75	0.60-0.90	0.040	0.050	0.15-0.25
3	0.90-1.03	0.30-0.50	0.025	0.025	0.15-0.25

3.2.1 Product analysis tolerances. Product analysis tolerances shall be in accordance with Fed. Std. No. 66.

3.3 Mechanical properties.

3.3.1 Hardness. The steel strip shall meet the hardness requirements shown in table II.

TABLE II. Hardness

Composition	Rockwell hardness 15 T scale
1	88 max.
2	88 max.
3	88-91

3.3.2 Bending. Steel strip of compositions 1 and 2 in the annealed condition shall withstand being bent at room temperature without cracking flat on itself longitudinally and transversely through an angle of 180 degrees around a radius equal to the thickness of the strip.

3.4 Decarburization. Depth of total and partial decarburization of the steel shall not exceed 0.001 inch (0.025 mm) or 3 percent of the thickness, whichever is greater (see 6.3).

3.5 Metallographic. Composition 3 shall show a fully spheroidized structure free from lamellar pearlite.

3.6 Dimensions and tolerances. Steel strip shall meet the dimensions specified in the invitation for bids, contract or order (see 6.2). Tolerances on specified sizes shall be as follows.

3.6.1 Thickness. Unless otherwise specified (see 6.2), the strip shall not vary in thickness more than the amounts shown in table III.

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Table III. Thickness tolerance

Ordered thickness inch (mm)	Thickness tolerances, plus or minus, inch (mm)
Up to 0.012 (0.30), incl.	0.0003 (0.0076)
Over 0.012 (0.30) to 0.020 (0.51), incl.	.0005 (0.0127)
Over 0.020 (0.51) to 0.025 (0.64), incl.	.0007 (0.0178)
Over 0.025 (0.64) to 0.030 (0.76), incl.	.0010 (0.0254)

3.6.2 Width. Unless otherwise specified (see 6.2), the strip shall not vary from the specified width by more than plus or minus 0.005 inch (0.127 mm) for widths up to and including 9 inches (229 mm).

3.6.3 Cross-bow. Tolerances on cross-bow shall be negotiated between the contractor and the procuring activity.

3.6.4 Camber. The edgewise curvature (camber) of the strip shall not exceed 1/4 inch (6 mm) in any 8-foot (2.4 m) portion of total length. No reverse camber shall be permitted.

3.6.5 Coils.

3.6.5.1 Arbor diameter. The arbor diameter of coils shall be approximately 16 to 20 inches (406 to 508 mm).

3.6.5.2 Weight. Coils shall weigh approximately 200 pounds (91 kg) per inch (25.4 mm) of width maximum.

3.6.5.3 Number of pieces. Coils shall contain not more than two pieces of strip, each of which shall be over 10 feet (3 m) in length.

3.7 Finish. Unless otherwise specified (see 6.2), strip shall be furnished with No. 2 (Regular Bright) finish.

3.8 Edge. Unless otherwise specified (see 6.2), strip shall be furnished with a No. 3 slit edge.

3.9 Identification marking. Each coil shall be marked in accordance with Fed. Std. No. 183.

3.10 Workmanship. The steel shall be clean and free from imperfections such as laminations, segregation, and surface defects which due to their nature, degree or extent, adversely affect the material for the intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified (see 6.2), a lot shall consist of not more than 1,000 pounds (450 kg) of strip of the same heat, composition, thickness and width subjected to inspection at the same time.

4.2.1 Product analysis. A lot for product analysis of chemical composition shall consist of one heat.

4.3 Sampling.

4.3.1 Chemical analysis. One sample shall be taken from each lot for product analysis. Samples shall be taken in accordance with method 111 or 112 of Fed. Test Method Std. No. 151.

4.3.2 Mechanical properties.

4.3.2.1 Hardness. Four samples shall be taken from each lot. The samples shall represent each end of two different coils selected randomly.

4.3.2.2 Bending. Samples for bend testing shall be taken from two different coils in each lot. Each sample shall be of sufficient size to obtain one longitudinal and one transverse test specimen therefrom.

4.3.2.3 Decarburization. Two samples shall be taken from each lot representing two different coils.

4.3.2.4 Metallographic. For composition 3, the samples selected for decarburization shall be used for determination of metallographic structure.

4.4 Examination.

4.4.1 Visual. A representative number of coils in each lot shall be examined for conformance with the requirements for edge (see 3.8), finish (see 3.7), identification marking (see 3.9) and workmanship (see 3.10).

4.4.2 Dimensional. A representative number of measurements shall be taken on each lot to determine conformance with the requirements for dimensions and tolerances (see 3.6) and coil (see 3.6.5).

4.4.2.1 Camber. Camber is the deviation of a side edge from a straight line. The deviation shall be obtained by placing an 8-foot (2.4 m) straight edge on the concave side and measuring the maximum distance between the steel edge and the straight edge.

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4.4.3 Packaging inspection. Prior to shipment, examination shall be made to determine conformance with the requirements of section 5.

4.5 Tests.

4.5.1 Test specimens.

4.5.1.1 Chemical composition. Test samples and specimens shall be prepared in accordance with method 111 or 112 of Fed. Test Method Std. No. 151, as applicable.

4.5.1.2 Hardness. Specimens shall be prepared in accordance with ASTM E 18.

4.5.1.3 Bending. One longitudinal and one transverse specimen shall be prepared in accordance with ASTM E 290.

4.5.1.4 Decarburization (see 6.3) and metallographic. Two specimens shall be prepared from each sample. The prepared surface of the specimen shall be not less than 1 inch (25.4 mm) in length representing the full thickness and shall be perpendicular to the direction of rolling. Specimens shall be prepared using precautions to ensure freedom from disturbed metal (see 6.3).

4.5.2 Test procedures.

4.5.2.1 Chemical composition. Chemical analysis shall be conducted in accordance with method 111 or 112 of Fed. Test Method Std. No. 151. Referee analysis shall be by method 111.

4.5.2.2 Hardness. Hardness tests shall be conducted in accordance with ASTM E 18.

4.5.2.3 Bending. Cold bending tests shall be conducted in accordance with ASTM E 290.

4.5.2.4 Decarburization (see 6.3) and metallographic. Microexamination of specimens shall be made at a magnification of 100 diameters.

4.6 Rejection.

4.6.1 Examination. If the sampling for visual, dimensional, or packaging fails to meet the requirements of this specification when examined in accordance with 4.4, the lot shall be rejected.

4.6.2 Tests. A lot shall be rejected for failure to comply with any of the specified requirements when tested in accordance with 4.5.

4.6.3 Retests. Retests shall be permitted in accordance with Fed. Test Method Std. No. 151.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-STD-163 for the levels or degree of protection specified (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order (see 6.2), shipments shall be marked in accordance with the requirements of MIL-STD-163.

5.3 Marking for delivery. In addition to marking specified in the contract or order (see 6.2), shipments shall be marked in accordance with MIL-STD-163.

6. NOTES

6.1 Intended use. The steel strip covered by this specification is for a highly specialized application in timing movements and it is essential that the steel be homogeneous and uniform in thickness and quality throughout the coil.

6.1.1 Composition 1 is intended for use in mechanical time fuze escapement levers and timing disk springs.

6.1.2 Composition 2 is for mechanical time fuze hammer springs and adjusting guides.

6.1.3 Composition 3 is for mechanical time fuze disks.

6.1.4 Compositions 1 and 2 are heat treated by the purchaser. Composition 3 is used in the as received condition.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Composition (see 1.2.1 and 3.2).
(c) Destination of pilot sample (see 3.1).
(d) Dimensions (see 3.6).
(e) Width tolerances, if other than as specified herein (see 3.5.2).
(f) Finish, if other than No. 2 is required (see 3.7).
(g) Edge, if other than No. 3 slit edge is required (see 3.8).
(h) Lot size, if other than as specified in 4.2.
(i) Levels of packaging and packing desired (see section 5).
(j) When special marking for shipment requirements are necessary (see 5.2).

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6.3 Decarburization. Measurement of decarburization by microexamination is contained in Society of Automotive Engineers Recommended Practice J419, SAE Handbook 1963, Society of Automotive Engineers, 485 Lexington Avenue, New York, NY 10017.

Custodians:
Army - MR
Navy - OS

Preparing activity:
Army - MR
Project No. 9515-0394

Review activity:
Army - MI

User activity:
Army - AR

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